

01 June 2026

SQL Aggregates and Operators Assignment

1) List all distinct departments in the students table

```
SELECT DISTINCT department  
FROM Students;
```

Department
IT
HR
FINANCE

2) Get the average age of students per department

```
SELECT department, AVG(age) AS AVG_AGE  
FROM Students  
GROUP BY department;
```

Department	AVG_AGE
IT	20,5
HR	22
FINANCE	23

* 3) Get all students whose age is between
④ 21 and 23

(2)

```
SELECT student_id,  
       name,  
       age,  
       department  
FROM students  
WHERE age BETWEEN 21 AND 23 ;
```

Student-id	Name	Age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

3) Show departments with more than 1 student

```
SELECT department, COUNT(student_id) AS  
       Student Count  
FROM students  
GROUP BY department  
HAVING COUNT(student_id)
```

Department	Student-count
IT	2
HR	2

③

5 List all students in the IT or HR department who are older than 21

```
SELECT student-id,
       name,
       age,
       department
FROM students
WHERE Age > 21 AND department IN('IT',
                                   'HR');
```

Student-id	Name	Age	Department
2	BOB	22	HR
5	EVE	22	HR

6) Show the total credits per department, only for department with more than 5 total credits

```
SELECT department, SUM(credits) AS total-credits
FROM courses
GROUPED BY department
WHERE total-credits > 5
HAVING total-credits > 5;
```

Department	total-credits
IT	11

④

7) List all courses that do not have 4 credits

```
SELECT Course-id,
        Course-name,
        department,
        Credits
FROM Courses
WHERE Credits <> 4;
```

Course-id	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8) Show the top 3 courses by credits in descending order

```
SELECT Course-id,
        Course-name,
        Credits
FROM Courses
ORDER BY Credits DESC
LIMIT 3
```

Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

5

a) Get the maximum, minimum and average grade across all enrollments

```
SELECT MAX(grade) AS max-grade;
       MIN(grade) AS min-grade,
       AVG(grade) AS AVG-grade
FROM enrollments
```

MAX-Grade	MIN-grade	AVG-grade
90	78	84.6

b) Count how many enrollments exist per course

```
SELECT course-id, COUNT(enrollment-id) AS
       enrollment-count
FROM enrollments
GROUP BY course-id;
```

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

⑥

11) Find total salary and total ~~be~~ bonus per department

```
SELECT department,
SUM (total salary) AS total - salary,
SUM (total bonus) AS total - bonus
FROM Salaries
GROUP BY Department
```

Department	total - salary	total - bonus
IT	122 000	105 000
HR	109 000	75 000
Finance	70 000	6 000

12) Show departments where Average salary is above 55 000

```
SELECT department,
AVG (Salary) AS Average - Salary
FROM Salaries
GROUP BY Departments
HAVING Average - salary > 55 000
```

Department	Average - salary
IT	61 000
Finance	70 000

13) List employees whose salary plus bonus is ≥ 60000

```
SELECT employee-id,  
        Name,  
        Salary,  
        Bonus  
        SUM (Salary + Bonus) AS total-compensation  
FROM Salaries  
WHERE total-compensation  $\geq$  60 000
```

employee-id.	Name	Salary	Bonus	total-compensation
1	Tou	60 000	5 000	65 000
23	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

14) Show total and average budget per department.
Only include departments with average budget above 70 000

```
SELECT department  
        SUM(Budget) AS total-budget  
        AVG (Budget) AS avg-budget  
FROM projects  
GROUP BY department  
HAVING Avg-budget  $>$  70 000
```

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

15) List all projects with budgets ^{between} ~~per department~~ 50 000 and 120 000 excluding marketing

```
SELECT project_id,  
       project_name,  
       department,  
       budget  
FROM   Projects  
WHERE  budget Between 50 000 and 120 000  
AND    department <> 'Marketing';
```

project_id	project_name	Department	Budget
1	AI app	IT	120000
2	Payroll System	FINANCE	80000
35	HR Portal	HR	50 000